



# INTRODUCTION TO BIOLOGY



## DEFINITIONS

**BIOLOGY:** Biology is the branch of science which deals with the study of living things.

**CELL:** Cell is the basic structural and functional unit of living organism.

**PROTOPLASM:** It is the combination of organic and specific inorganic substances.

## SHORT/DETAILED ANSWER & QUESTIONS

**Q.1 What is Biology? How many Divisions and Branches of Biology? Name them and explain briefly?**

Ans. **BIOLOGY:**

The word Biology is composed of two Greek words "Bio" meaning "Life" and "Logos" meaning "Discourse", thought or reasoning or in simple study. It is the branch of natural science that deals with organisms and study different phenomena of life.

**DIVISION OF BIOLOGY:**

There are three main divisions of Biology

i. Zoology

ii. Botany

iii. Microbiology

**ZOOLOGY:**

The word zoology is derived from Greek Language "Zoon" meaning "animals" and "Logos" meaning "Study or Knowledge". It deals with the study of animals.

**BOTANY:**

The word is taken from Greek Language "Butane" meaning "Plants" and "Logos" meaning "Study or Knowledge". It deals with the study of plants.

**MICROBIOLOGY:**

It Deals with the study of microscopic organisms such as bacteria, Viruses etc. which can be seen only with the help of microscope.

**BRANCHES OF BIOLOGY:**

Some Important branches of Biology are defined below.

|                  |                          |                          |
|------------------|--------------------------|--------------------------|
| 1. Morphology    | 2. Anatomy               | 3. Cell Biology          |
| 4. Histology     | 5. Physiology            | 6. Taxonomy              |
| 7. Genetics      | 8. Developmental Biology | 9. Environmental Biology |
| 10. Paleontology | 11. Biotechnology        | 12. Socio-biology        |
| 13. Parasitology | 14. Pharmacology         | 15. Molecular biology    |

**MOROPHOLOGY:**

The study of external form and structure of organisms.

**ANATOMY:**

The study of external parts of body of living organisms by cutting them open.

**CELL BIOLOGY:**

The study of cells and its organelles.

**HISTOLOGY:**

The study of structure of tissues of plants and animals.

**PHYSIOLOGY:**

The study about functions of living organisms.

**TAXONOMY:**

The study of the rules, principles, grouping and naming the living organisms.



**GENETICS:**

The study of heredity that is transferring of characters from parents to offspring.

**EMBRYOLOGY: (Developmental Biology)**

The study of formation and development of embryo.

**ENVIRONMENTAL BIOLOGY:**

The study of relationship between living organisms and non-living factors of environment and their factors on each other.

**PALEONTOLOGY:**

The study of remote past organic life, with the help of fossils.

**BIOTECHNOLOGY:**

The study about techniques for manipulation of gene to bring the changes in structure and location of genes to achieve desirable characters is called biotechnology.

**SOCIO-BIOLOGY:**

The study of social behavior of living organisms i.e., interaction between themselves.

**PARASITOLOGY:**

The study of parasites.

**PHARMACOLOGY:**

The study about action of drugs.

**MOLECULAR BIOLOGY:**

The study of organic molecules which constituent cell and its organelles.

**Q. Describe the relationship of Biology to other sciences.**

Ans. **RELATIONSHIP OF BIOLOGY WITH OTHER SCIENCES**

Biology is considered as interdisciplinary science, which is related with other sciences, some of these are mentioned below.

**BIOPHYSICS:**

It is a branch of physics, in which laws and techniques of physics are applied to explain the process of life.

**BIOMETRY: (BIOMECHANICS)**

The branch of mathematics which collects data of living organisms. It plays very important role in research.

**BIOCHEMISTRY:**

It is a branch of biology which deals with the study of molecules which form living organisms or cell and requires authentic knowledge about biology and chemistry to explain the synthesis to biomolecules and function of different molecules in the body of organisms.

**BIOGEOGRAPHY:**

It deals with the distribution of different living organisms in different geographical regions of the world. Many living organisms are restricted to particular geographical regions due to environmental conditions.

**BIO-ECONOMICS:**

This deals with the economically important organisms involved in production e.g., meat production etc. are calculated for cost value and profit value.



## Q. What does Quran describe about life?

Ans. QURANIC TEACHING ABOUT ORIGIN OF LIFE:

أَوَلَمْ يَرِ الَّذِينَ كَفَرُوا أَنَّ السَّمَوَاتِ وَالْأَرْضَ كَانَتَا رَتْقًا فَفَتَقْنَاهُمَا وَجَعَلْنَا مِنَ الْمَاءِ كُلَّ شَيْءٍ حَيٍّ أَفَلَا يُؤْمِنُونَ

ترجمہ

کیا کافروں نے یہ نبیال نہ کیا کہ آسمان اور زمین ملے ہوئے تھے تو ہم نے انہیں کھول دیا اور ہم نے ہر جاندار چیز پانی سے بنائی تو کیا وہ ایمان نہیں لائیں گے؟

"We made everything of water. Will they not than believe"

(Surah AL-Anbiya, Ayah 30)

وَاللَّهُ خَلَقَ كُلَّ دَابَّةٍ مِنْ مَّاءٍ فَمِنْهُمْ مَنْ يَمْشِي عَلَى بَطْنِهِ وَمِنْهُمْ مَّنْ يَمْشِي عَلَى رِجْلَيْنِ وَمِنْهُمْ مَّنْ يَمْشِي عَلَى أَرْبَعٍ يَخْلُقُ اللَّهُ مَا يَشَاءُ إِنَّ اللَّهَ عَلَى كُلِّ شَيْءٍ قَدِيرٌ

ترجمہ

اور اللہ نے زمین پر چلنے والا ہر جاندار پانی سے بنایا تو ان میں کوئی اپنے پیٹ کے بل چلتا ہے اور ان میں کوئی دو پاؤں پر چلتا ہے اور ان میں کوئی چار پاؤں پر چلتا ہے۔ اللہ جو چاہتا ہے پیدا فرماتا ہے۔ بیشک اللہ ہر شے پر قادر ہے۔

"And Allah has created every animal from water of them there are some that creep on their bellies, some that walk on two legs; and some that walk on four. Allah creates what He will Lo! Allah is able to do all"

(Surah AL-Nur, Ayah 45)

## VIEW ABOUT IMPORTANCE OF PLANTS AND ANIMALS:

وَفِي الْأَرْضِ قِطْعٌ مُّتَجَوِّرَاتٌ وَجَنَّتْ مِنْ أَعْنَابٍ وَزَيْتُونٌ وَنَخِيلٌ صِنَوَانٌ وَعَذِيرٌ صِنَوَانٌ يُسْقَى بِسَاءٍ وَاحِدٍ وَنُقْضِلُ بَعْضَهَا عَلَى بَعْضٍ فِي الْأَكْلِ إِنَّ فِي ذَلِكَ لَآيَاتٍ لِّقَوْمٍ يَعْقِلُونَ

ترجمہ

اور زمین کے مختلف حصے ہیں جو ایک دوسرے کے قریب قریب ہیں اور انگوروں کے باغ ہیں اور کھیتی اور کھجور کے درخت ہیں ایک جڑ سے اگے ہوئے اور الگ الگ اگے ہوئے، سب کو ایک ہی پانی دیا جاتا ہے اور پھلوں میں ہم ایک کو دوسرے سے بہتر بناتے ہیں، بیشک اس میں عقل مندوں کے لیے نشانیاں ہیں۔

"And in the earth are neighboring tracts, vineyards and ploughed lands, and datepalms, like and unlike which are watered with one water. And we have made some of them excel others in fruits. Lo! Herein verily are portents for people who have sense."

(Surah AL Ra'd Ayah 4)

## Q. Describe the contribution of different Muslim scientist in the field of Biology.

Ans. Muslim scientist made great contribution to the science and we are aware of their success in different field of science.

**JABIR BIN HAYAN (721-815 A.D.):**

1. He was born in Iran and practiced in Iraq.
2. He introduced experimental investigation in chemistry and also wrote a number of books on plants and animals.
3. His famous books are Al-Nabatat and Al-Haywan.

**ABDUL MALIK ASMAI (740-828 A.D.):**

1. He is considered the first Muslim scientist who studied animals in detail.
2. His famous writing books included Al-Ibil (camel), Al-Kheil (horse), Al-Wahoosh (animals) and Khalaaqal-Insan.



| SCIENTIST NAME                     | BOOKS                                              |
|------------------------------------|----------------------------------------------------|
| JABIR BIN HAYAN<br>(722-817 A.D)   | Al-Nabatiat<br>(on plants)                         |
| ABDUL MALIK<br>ASMAI (742-828 A.D) | Al-Kheil, Al-Ibil                                  |
| BU ALI SINA (980 A.D)              | Khalaqal-Insan,<br>Al-Qanoon , Fil Tib Al<br>Shafa |

### Q. Describe the level of organization.

Ans. **THE LEVEL OF ORGANIZATION:**

The levels of organization in living world are based on chemical foundation. All the living organisms are made up of cells and the protoplasm of cell is the physical as well as chemical basis of life. These levels are as follow:

#### **ATOMIC LEVEL OF ORGANIZATION:**

All the matter is made up of elements, which is composed of atom. Each atom is made up of sub-atomic particles, such as electrons, protons and neutrons. In nature, there are more than 100 kinds of elements and among these 16 elements are called as bio-elements, which are vital for life. Only six elements such as C, H, O, N, S and P are called basic elements of life.

#### **MOLECULAR LEVEL OF ORGANIZATION:**

Molecules are formed by the binding of atoms. These organic molecules of cells are called as bio-molecules. These are constructed in great variety and complexity. They are classified as micro-molecules and macro-molecules. Glucose, amino acid and fatty acids are micro-molecules, whereas carbohydrates, proteins and lipids are macro-molecules. The units of macro-molecules combine together to form macromolecules.

#### **CELLULAR LEVEL OF ORGANIZATION:**

The biomolecules when work together in the form of suspension, it is called Protoplasm. Protoplasm is the combination of organic and specific inorganic substances. When protoplasm works in the form of a unit, this is called cell. Cell is the basic unit of living organisms when similar types of cells organize together in a group, they are called tissues. The different types of tissues arranged in a particular manner to work together are called organs.

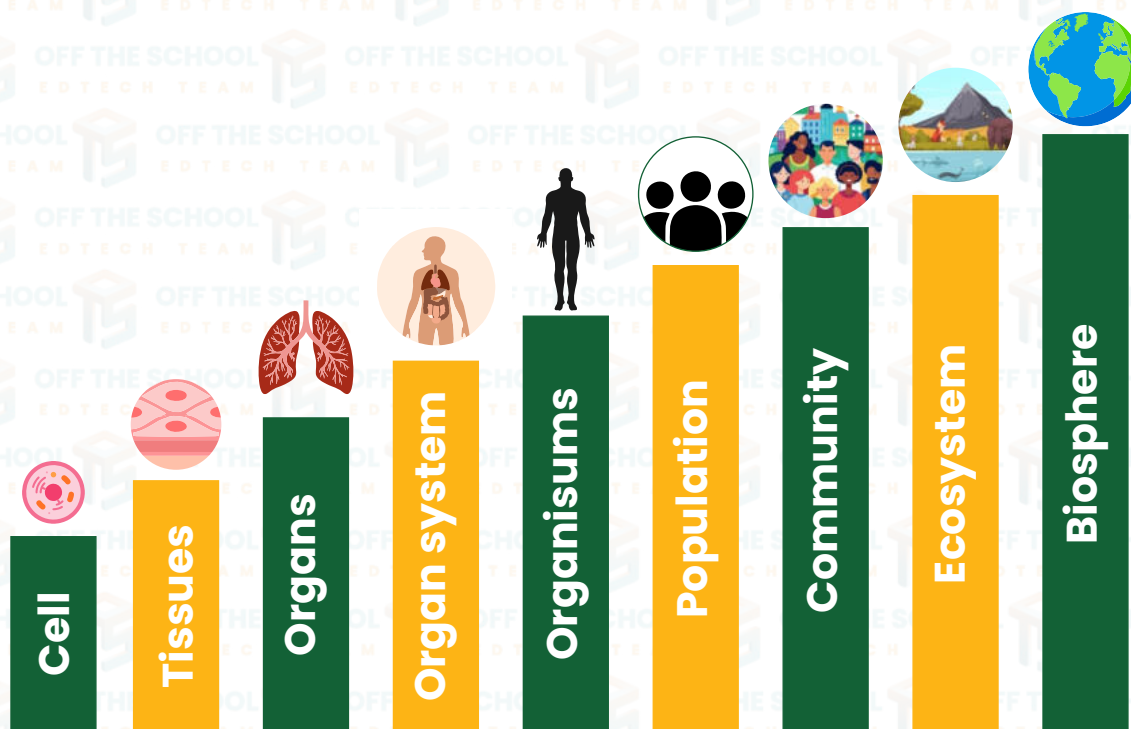
Organs of different types work in a coordinated manner to perform a function which is called organ-system. When different organ-systems function in co-ordination as a unit, they form a body or Multicellular Organism.



**“BOOKS ARE GOOD TO SHOW THE THEORY OF THINGS, BUT DOING THE THING ITSELF IS WHAT COUNTS.”**

—THOMAS A EDISON—





### TAXONOMIC LEVEL:

There is another level of organization which is related with living organisms. The species is the smallest unit of taxonomic level of organization, which includes morphologically similar living organisms which inter-breed and produce fertile offspring.

### POPULATION LEVEL:

All the members of a species, living in specific habitat are called Population. A group of parrots living on tree, is called parrot population.

### COMMUNITY LEVEL:

The members of different species, living in specific habitat are called as Community. A group of different kinds of birds, living on tree, is called as birds, living on tree, is called as bird community.

### ECOLOGICAL SYSTEM:

Communities always depends upon their non-living environment is a reciprocal interaction for their survival. For example, oxygen for respiration is obtained from environment and in turn for respiration is obtained from environment and in turn given out CO<sub>2</sub>. This interaction is called Ecosystem for Ecological system.

### BIOSPHERE LEVEL:

The part of earth where life exists is called biosphere. It consists of different kinds of eco systems.

**Q. What is cell organization? How many types of cell? Explain with the help of example.**

Ans. **CELL ORGANIZATION:**

In the living organisms the cell organizes in three ways of to make bodies of organisms. cell make unicellular, colonial and multicellular organizations and the other organisms formed through these organization are unicellular organism, colonial organisms and multicellular organisms respectively.

### 1. UNICELLULAR ORGANIZATION:

In unicellular organisms only one cell makes the life. All the life activities are carried out by the only cell.

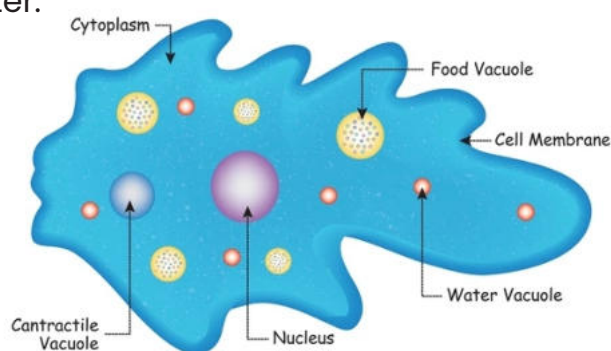
### FOR EXAMPLE:

Amoeba, Paramecium and Euglena



**AMOEBA:**

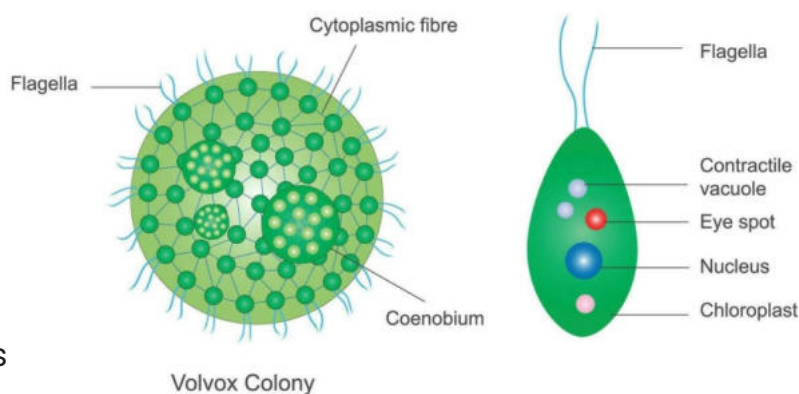
- Amoeba is a unicellular organism found in the mid shallow pond, pools and at the stagnant water.
- Its size is about 0.25 millimeters.
- It has a regular shape.
- It has a cell membrane which helps in movement of molecules and protect cytoplasm.
- The outer part of cytoplasm is clear and transparent called ectoplasm (gel) and inner part of cytoplasm is called endoplasm (sol).
- The cytoplasm contain nucleus, food vacuoles, mitochondria etc. Amoeba moves by false foot, called pseudopodia.

**Amoeba Structure****2. COLONIAL ORGANIZATION**

In colonial type of cellular organization, many unicellular organisms live together but do not have any divisions of labour among them each unicellular organism is a colony live. Its own life and does not depend upon on other cells for its vital requirements. Volvox is a green alga found in water is an example of colonial organization. Hundreds of volvox cells make a colony.

**FOR EXAMPLE:****VOLVOX:**

- Volvox once called algae that live together in a colony.
- Each volvox cell has two flagella.
- The flagella beat together to roll the body in water.
- Volvox cells have chlorophyll and make their own food by photosynthesis.
- Volvox are not harmful to humans because they do not produce any toxic substances.

**3. MULTICELLULAR ORGANIZATION**

In multicellular organization, cells can organize in tissues, organs and organ system.

**FOR EXAMPLE:****MUSTARD PLANT:**

SCIENTIFIC NAME: *Brassica campestris*

**1. SOWING:**

It is sown in winter and produces seed at the end of winter.

**2. IMPORTANCE:**

Its plant body is used as vegetables and its seeds are used for extracting oil.

**3. BODY PARTS:**

Plant body consists of two parts.

**4. VEGETATIVE PARTS:**

It includes roots, branches and leaves. These do not take part in sexual reproduction.



**SCIENTIFIC REASONS:****Q.1. Why subject biology is named as multidimensional subject?**

Ans: Biology is a multidimensional subject that links with sciences. It is an interdisciplinary science, which is related with other sciences such as biometry, biochemistry or biotechnology etc.

**Q.2. How farming profession helps mankind?**

Ans. The profession of farming deals with the development and maintenance of different types of farm. e.g., animal farm, poultry farm, fruit farm etc. In some farms animal breeding technologies are used for the production of animals which are better protein and milk source. In poultry farm chicken and eggs are produced. While in fruit farms different fruit yielding plants are grown.

**Q.3. Why species is called as smallest taxonomic level?**

Ans: Specie is the smallest unit of taxonomic level of organization because it includes morphologically similar living organisms which inter-breed and produce fertile offspring.

**Q.4. How population is different from community?**

Ans. Member of a same species, living in specific habitat are called population such as a group of parrots living on a tree is called population while a group of different bird living on a tree is called bird community.

